

PAVE Coding Challenge (v0.8)

The purpose of this coding challenge is:

- To see what kind of coder you are
- So you can see what kind of technology you'll be working on
- So we can work together to solve a tricky problem and see if our communications skills are compatible.

When you've completed your solution, send us your github and we'll schedule a call to discuss your solution.

Your challenge is to build a fees API in `encore` (`encore.dev`) that uses a temporal (`temporal.io`) workflow started at the beginning of a fee period, and allows for the progressive accrual of fees. At the end of the billing period, the total invoice and bill summation should be available.

Requirements:

1. Able to create new bill
2. Able to add line item to an existing open bill
3. Able to close an active bill
 - a. indicate total amount being charged
 - b. indicate all line item being charged
4. Reject line item addition if bill is closed (bill already charged)
5. Able to query open and closed bill
6. Able to handle different types of currency, for the sake of simplicity, assume GEL and USD.

You are free to design the RESTful API that will be used by other services. The above requirements are not exhaustive, you're free to add in requirements that you think are helpful in making the Fees API more flexible and complete.

Your solution should:

- Use `encore` and `temporalite` (a local temporal)
- Use temporal signals
- Include a `Readme`
- Have unit tests

We treat Temporal as a first class store of state. You should model the monthly bill as a single workflow. For each of our customers, each month we will start a bill and add items to the bill before finally signalling that the end of the month has been reached and we now want to collect the money.

Things to consider.

- What is the correct type to store money? Int? BigInt? Float? String? What are the advantages to each?
- What are the correct semantics for the API?

We encourage you to use AI. In particular we suggest asking GPT-5 or Claude for a “critical code review” of your solution before you return it to us. This models how we work in the real world, we want to test your architectural skills, not nitpick small details that AI can help you with.

Hint: <https://encore.dev/docs/how-to/temporal>

Hint: you should use the Temporalite version to make it easy

<https://github.com/temporalio/temporalite>